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Pre-Lab #2 Report:

1. Make a list of anything missing from your lab kit. In most cases, the answer will be "Nothing."

Nothing missing from lab kit

2. Read the data sheets on the LTE-5208A Infrared LED and the LTR-4206E Phototransistor.
Explain how you will identify each pin on the physical devices.

On the LTE-5208A Infrared LED, the cathode is marked with a dot/indentation on one of the sides above the lead. On the LTR-4206E Phototransistor, the collector is marked with a flat side.

3. Determine the angle of emission for most of the light from the LTE-5208A Infrared LED.

At 0 degrees (hit the sensor straight on), there is a radial intensity of 1, which is the max value.

4. Determine how large a circle will be illuminated by the LTE-5208A at 80cm.

A circle with a radius of 67.127 cm will be illuminated by the LTE-5208A

5. Review the documentation for the LM324 op amp. What is the likely maximum output voltage when the power supply is 5V?
LM324 gain = 100db
20log(Vout/(Vin-1.5)) = 100dB - > log(Vout/3.5V) = 5 -> Vout/3.5 = 10^5 -> Vout=350kV
BUT it saturates at the power rail of 5V. So the answer is 5V.

If referring to the Voltage Follower Circuit specifically, the gain is 1, and thus Vout=Vin so Vout =3.5V.

6. Square waves are not really square. Include a selfie of your team from your planning meeting along with something that is square (or at least close enough to look square).